Big Data Tools and Emerging Technologies Paper

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Big Data Tools

Big data-specific tools are important because traditional data processing methods can struggle when datasets become too large, complex, or constantly updated. Tools such as Apache Spark, Hadoop, and Pandas can help organizations manage and analyze information more efficiently. Hadoop supports distributed storage and processing across multiple computers, while Spark can process large datasets using distributed computing. Pandas is useful for organizing, cleaning, and analyzing smaller datasets using DataFrames. These tools can also work together. For example, an organization could store large amounts of data using Hadoop, process the information with Spark, and use Pandas to examine a smaller portion of the results. This allows organizations to work with large amounts of information while still using familiar Python tools for analysis.

During the Module Five uCertify labs, I worked with Spark DataFrames and practiced operations such as reading CSV files, selecting columns, renaming columns, and displaying specific records. These activities helped me understand how Spark organizes information into rows and columns while providing tools to process much larger datasets. Spark can also be used to read data from different sources and parse text by separating or extracting useful information from larger collections of data. Organizations can use these capabilities for descriptive analysis, identifying trends, and preparing information for machine learning. For example, a business could analyze customer purchases to identify popular products and eventually use historical purchasing patterns to predict future demand (Apache Software Foundation, n.d.).

Relational vs. Non-relational Databases

Non-relational databases, also known as NoSQL databases, are useful for handling large amounts of data that do not always fit into traditional tables. Unlike relational databases, which organize information into rows and columns with defined relationships, non-relational databases can store information in different formats, such as documents, key-value pairs, and graphs. Both types of databases can work together within an organization. For example, a business might use a relational database to store customer orders and payment records while using a non-relational database to collect product reviews, website activity, or information from connected devices.

The variety of data collected through the internet of things (IoT) can also influence which tools an organization chooses. Devices such as smartwatches, security cameras, and temperature sensors collect different types of information, including numerical measurements, images, videos, and location data. A relational database may work well for organized records, while a non-relational database may be more practical for large amounts of changing or less structured information. Choosing the appropriate tools helps organizations manage their data more efficiently and makes it easier to analyze information from multiple sources (Amazon Web Services, n.d.-a).

Effects of IoT

The internet of things has become a major part of everyday life because so many devices can now connect to the internet and exchange information. This can benefit consumers by making everyday tasks more convenient while also helping businesses better understand how their products and services are being used. One example is a smart thermostat that learns a household's temperature preferences and automatically adjusts the heating or cooling system. This can make the home more comfortable while potentially reducing unnecessary energy use.

However, connected devices also raise privacy concerns because they may collect information about people's daily routines and habits.

Businesses can also use IoT devices to improve customer experiences. For example, a grocery store could use connected sensors to monitor refrigerator temperatures and receive alerts when equipment starts having problems. This would allow employees to respond before food spoils, helping maintain product quality and reducing waste. The information collected over time could also help the business identify equipment that frequently needs maintenance and predict when repairs may be necessary. This shows how IoT can benefit both the organization and its customers by improving efficiency, reliability, and overall service (Amazon Web Services, n.d.-c).

Machine Learning and AI

Artificial intelligence (AI) is a broad field focused on developing technology that can perform tasks associated with human intelligence, such as understanding language, solving problems, and making decisions. Machine learning is a part of AI that allows computers to recognize patterns and improve their performance by learning from data rather than relying entirely on explicitly programmed instructions. Although the two are closely related, not every AI system uses machine learning. Machine learning is especially useful when large amounts of information are available because it can identify patterns that might be difficult for people to recognize on their own.

Machine learning can improve consumer devices by helping them adapt to individual preferences and behaviors. For example, a smartwatch can analyze activity patterns and provide personalized fitness recommendations, while a smartphone can learn frequently used words to improve text predictions. AI can expand these capabilities even further through features such as voice assistants, language translation, image recognition, and accessibility tools. These technologies can make devices more convenient and useful, although their effectiveness still depends on the quality of the information they receive and how accurately they interpret it (Amazon Web Services, n.d.-b).

Impacts of AI

AI has had a significant impact on society by making information easier to access, automating repetitive tasks, and improving the way organizations analyze data. However, these benefits also come with concerns involving privacy, misinformation, bias, and cybersecurity. Since AI systems often depend on large amounts of information, organizations need to be careful about what they collect and how it is stored. Sensitive information could be exposed through data breaches, unauthorized access, or poorly secured systems. Organizations can reduce these risks by limiting unnecessary data collection, encrypting sensitive information, restricting access, and regularly reviewing their security practices.

Anonymization is another important consideration when organizations collect information for AI. This involves removing or changing identifying details so that data cannot easily be connected to a specific person. For example, a healthcare organization might use appropriately anonymized patient information to study patterns in medical conditions without directly identifying individual patients. This can be useful when analyzing large datasets for

research or improving services. However, removing names alone does not guarantee privacy because individuals may still be identified when different pieces of information are combined. Organizations should evaluate the risk of reidentification and use additional safeguards when working with sensitive information (National Institute of Standards and Technology, 2026).

AI and Business Strategies

AI can influence business strategies by helping organizations understand customer behavior, improve efficiency, and make more informed decisions. For example, a retail business could analyze purchasing patterns to determine which products are becoming more popular and adjust its inventory before demand increases. AI can also help businesses identify trends in customer feedback, forecast sales, and automate repetitive administrative tasks. However, businesses still need employees to evaluate the results and make decisions that consider situations the technology may not fully understand. Duranton (2019) emphasizes the importance of combining AI with human knowledge and judgment rather than relying entirely on automated decisions.

AI can also improve interactions between businesses and customers through tools such as virtual assistants and customer service chatbots. These systems can answer common questions, help customers find products, and provide basic support without requiring someone to wait for an available employee. For example, an online store could use an AI assistant to help customers track orders or find information about returns. This can make customer service more accessible while allowing employees to focus on complicated situations that require personal attention. I think the best approach is for businesses to use AI as a way to support their employees and customers rather than completely replacing human interaction.

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